KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

END OF YEAR III SEMESTER I SUPPLEMENTARY EXAMINATION
FOR
DIPLOMA IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: CLINICAL ENGINEERING I

DATE: 11th September 2023	**TIME: 2 HOURS (2:00PM – 4:00PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (20 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (2 Long Answer Questions)
2. Attempt **ALL** Questions
3. Write your **COLLEGE NUMBER** on all the answer sheets provided.
4. Ensure that all examination answer scripts are handed in at the end **of the examination**
5. Ensure you sign the examination register provided

DO NOT WRITE ANYWHERE ON THIS QUESTION PAPER

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SECTION 1: MULTIPLE CHOICE QUESTIONS (20 MARKS)

1. The difference between the true value and actual measured by biomedical sensor is the?
   A. Range of measurement instrument
   B. Accuracy of the measurement instrument
   C. Precision of the measurement instrument
   D. Drift of the measurement instrument
2. Which of the following approaches is the most effective for determining the concentration of oxygen inhaled?
   A. With an oxygen analyzer
   B. With a pulse oximeter
   C. Check the liter-per-minute flow of oxygen
   D. Take blood gas measurements
3. A complete blood gas analyzer as designed has the following except?
   A. Vacuum system for aspiration and flushing services
   B. Humidification system for calibration
   C. Thermostatic control for PH variation
   D. Precision regulator for flow and pressure control
4. Which of the following principles are used by the majority of transducers in enzyme electrodes?
   A. Magnetic
   B. Optical
   C. Colorimetric
   D. Amperometric
5. What is the wavelength of infrared LED in pulse oximetry?
   A. 660nm
   B. 740nm
   C. 905nm
   D. 950nm
6. Which medical devices display the patient's heart's graphical interface?
   A. Stethoscope
   B. Phonocardiogram
   C. Laryngoscope
   D. Endoscope
7. When blood is tested for oxygen saturation in oximetry while flowing through the vascular system, the procedure is known as?
   A. In vivo oximetry
   B. Reflection oximetry
   C. In vitro oximetry
   D. Transmission oximetry
8. Blood glucose monitoring is a way of checking the concentration of glucose in blood using
   A. Glucometer
   B. Biosensor
   C. Non-enzymatic
   D. Polyacrylamide gel.

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9. Which of the following is the cell technique in hematology analyzers?
   A. Dark field
   B. Flow cytometry
   C. Ultrasound technique
   D. Doppler technique
10. Which of the following is not a measurement mode in biochemistry analyzer?
   A. Direct mode
   B. Indirect mode
   C. Urine mode
   D. Serum mode
11. The ultrasonic blood flow meters are based on which principle?
   A. Doppler and oscillatory
   B. Transit and oscillatory
   C. Doppler and piezoelectric
   D. Oscillatory and piezoelectric
12. The temperature measurement of a metal melting furnace (oven) which is related to interior temperature by certain factor is example of?
   A. Direct measurement
   B. Null measurement
   C. Indirect measurement
   D. Nominal measurement.
13. What is the primary purpose of biological safety cabinets?
   A. To protect the user and the sample
   B. To protect the sample and the environment
   C. To protect the user, sample and the environment
   D. To protect the user and the environment
14. Active transducers work on the principle of
   A. Energy conversion
   B. Mass conversion
   C. Energy alteration
   D. Volume conversion
15. Which one of the following is a representation of an absolute error?
   A. X ±Δcm
   B. X ±1%cm
   C. X ±cm
   D. ∇ +cm
16. The following are factors that influence pulse rate except?
   A. Age
   B. Gender
   C. Physical condition
   D. Weight
17. Which of the following is not part of white blood cell?
   A. Lymphocytes
   B. Granulocytes
   C. Monocytes
   D. Thrombocytes

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18. Type of the biological safety cabinet that protects the operator, the product, environment and suitable for working with agent classified biosafety level 4 is?
    A. Class II - Type BI
    B. Class II- Type A
    C. Class III
    D. Class II - Type B3
19. Which principle does electromagnetic blood flow meter work with?
    A. Coulomb's Law
    B. Law of magnetic induction
    C. Faraday's Law
    D. Gauss Law
20. Blood pressure is the force against
    A. Arterial wall
    B. Capillaries wall
    C. Veins wall
    D. Venacava

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

1. a) List two techniques for measuring the blood flow and velocity. (2marks)
   b) State three functions of blood flow meter. (3marks)
2. a) Define oximetry as used in blood gas determination and give its mathematical expression. (2marks)
   b) State any three applications areas of oximetry. (3marks)
3. a) Give two reasons why monitoring of blood glucose is important. (2marks)
   b) Mention three parameters measured using blood gas analyzers. (3marks)
4. a) State three equipments necessary to carry out Eliza test. (3marks)
   b) State two types of electromagnetic flow meters. (2 marks)
5. a) State three types of pulse sensors. (3marks)
   b) State two disadvantages of indirect method of blood pressure measurement. (2marks)
6. List five benefits of using optical fiber temperature sensor over traditional sensors. (5marks)
7. a) State two types of monochrometers. (2marks)
   b) State the three parameters that affect the PH measurement. (3marks)
8. Describe five artifacts in ultrasound blood flow measurement. (5marks)

SECTION 3: LONG ANSWER QUESTIONS (40 MARKS)

1. a) State three categories of parameters that sensors detect in biomedical measurement. (3marks)
   b) Explain the following terms used in biomedical measurement;
      i. Accuracy
      ii. Precision
      iii. Resolution
      iv. Reproducibility
      v. Sensitivity (10marks)

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c) State any four factors encountered in measurement. (4marks)
d) State the three categories of measurement. (3 marks)

2. a) Differentiate between respiratory rate and pulse rate. (2marks)
   b) Draw a block diagram of an infrared thermometer. (5marks)
   c) State any four maintenance routine of biological safety cabinet. (4marks)
   d) Differentiate between error and mistake giving out relevant examples. (2marks)
   e) Draw a diagram showing light path from the source to the photo sensor in Biochemistry analyzer. (7marks)

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